Design Process Overview

Poster: PDF · **Promotional material:** pitch script · **Prototype:** runs on a laptop and phones, no login; run steps for app-v2 and v0.3 are in the README. · **Team 25:** Aman Kushwaha, James Olbes, Anujitha Fernando, Kenan Chu

Summary

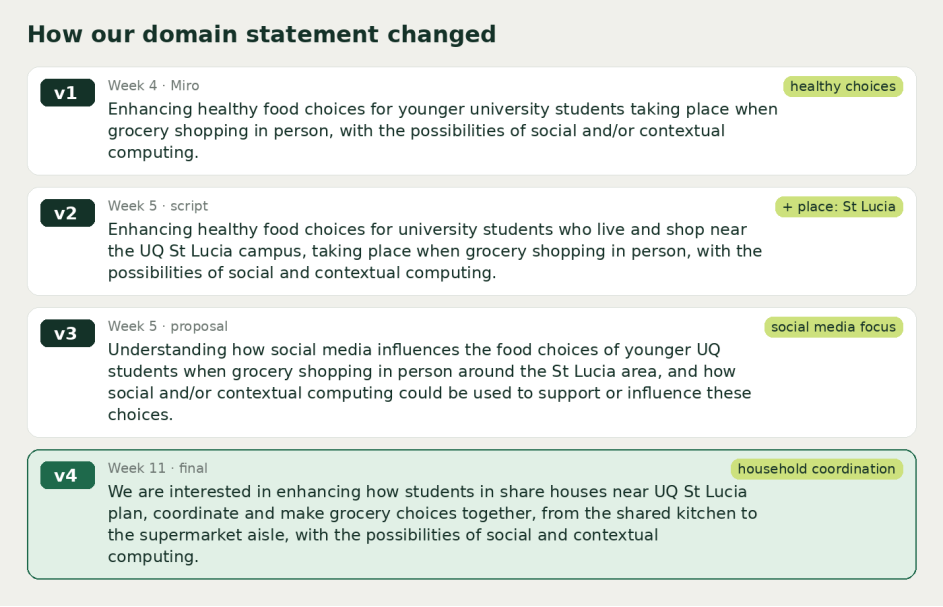
We investigated how students living together near UQ St Lucia plan, coordinate and make grocery decisions. Price decides most purchases, and “healthy” means something different to everyone. Most shopped alone, yet their shopping overlapped through shared staples, fridge space and lists sent to whoever was going. The useful social influence isn’t a feed of friends’ meals but the everyday coordination of a household: who needs what, who’s going, what’s already in the fridge, and who pays.

NutriCart Hub is our response. Housemates plan together at the kitchen Home Dock, and whoever shops takes the Cart Unit with the household's list, requests and budget. The social computing sits in six moments where several housemates take part (G1–G6).

Process

1. Understanding the domain (Weeks 3–6)

Our first domain statement was about "enhancing healthy food choices". Round 1 interviews asked for **stories** not opinions, and showed **price first** ("99 out of 100 times") and social influence that works through **trust and timing**: a housemate who finds the cheap milk, a reel seen right before going out. The literature showed food-choice tools are mostly individual (Marshall et al., 2022), and that social media inspiration rarely carries into the store (Wang et al., 2024). Proposal feedback asked: whose definition of healthy? Why an app?



Our domain statement went through four versions, from "healthy choices" and "social media" to household coordination.

2. An app as a probe, not a product (Weeks 6–7)

v1 was a social shopping app with friend recommendations, group goals, recipes and a checklist. We used it as a **design probe**. People wanted **prices**, not friends' recipes. Group goals divided them: one participant read a healthy-dinner goal as "essentially telling me that I'm fat".

3. Reframing and divergence (Weeks 7–8)

Thematic synthesis reframed the problem: *support real grocery decisions around price, time, personal priorities and household needs, without adding effort, judgement or another app.* We explored non-app concepts (a budget buzzer, a "grocery baton") and combined the strongest into **v2**: a cardboard Home Dock and detachable Cart Unit, tested using Wizard of Oz. Testing kept the **shared list, running budget and neutral compare**, and told us to shrink the Cart Unit, keep alerts rare ("noise, not help") and cut data entry ("will be a pain"). This gave us four requirements: inform without judging, put price first, add no extra effort, and support coordination the household already does.



Storyboard: Jay forgets what's at home and buys doubles; with the Hub, the household's needs travel with him. More concepts and mock-ups are in documents/design.

4. "What's the point of the social part?" (Week 9 Stand-up)

Staff said the social aspect was weak: apps are individual by nature, so we redesigned around **six group moments** and placed them on the time–space matrix:

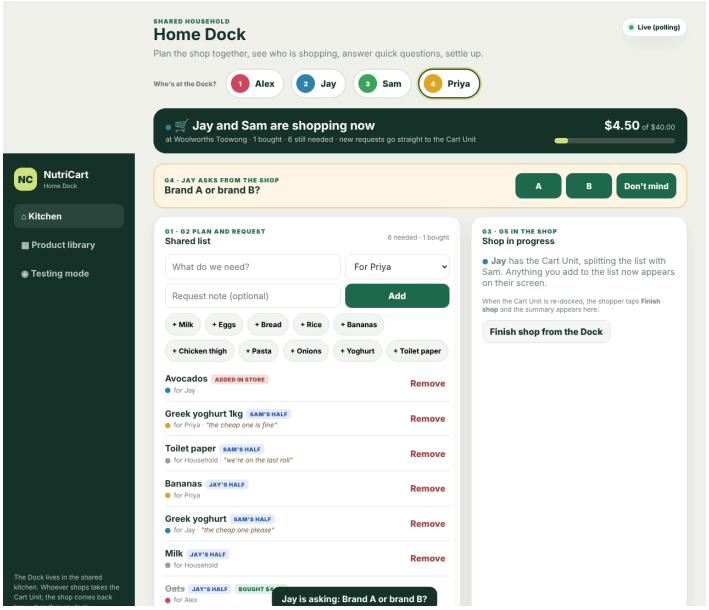
	Same time	**Different time**
Same place	G1 plan together at the Dock · G5 Split Shop in store	G2 leave a request · G3 hand over the baton (then travels with the shopper) · G6 return and settle
Different place	G4 one quick question home	–

The Hub now **facilitates** coordination households already do (Olsson et al., 2020), uses visible traces rather than notifications (Schmidt, 2002) and gives every housemate something back (Grudin, 1994).

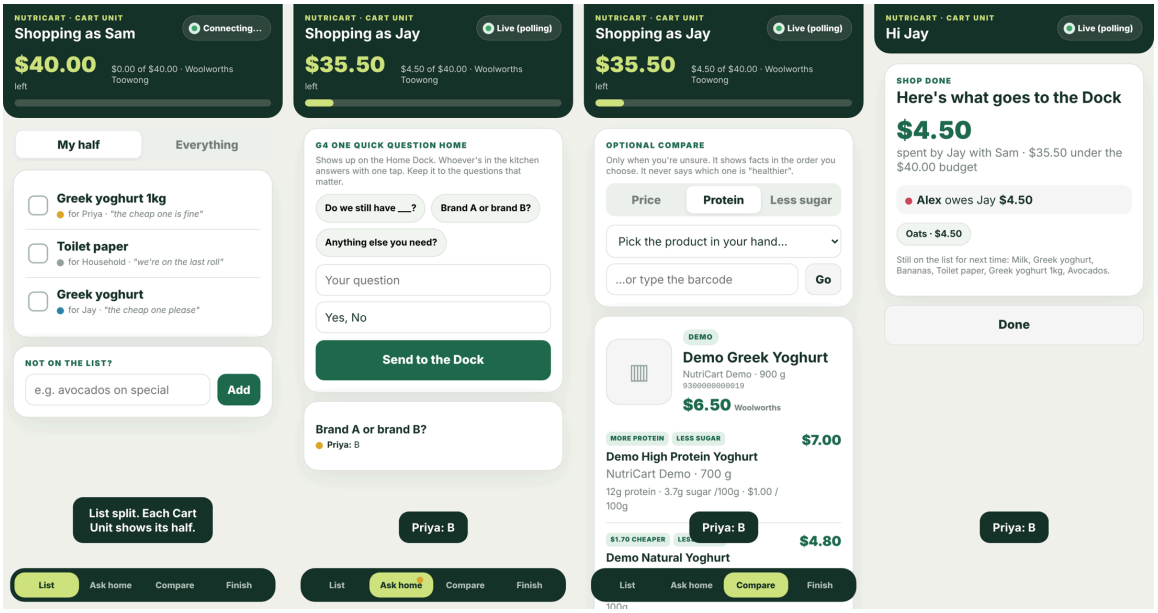
Applying the Designing for Location Dependence framework (Design for Location, n.d.) showed that decisions start at home, not at the store entrance, so NutriCart follows a home → store → home trajectory: the Dock belongs to the kitchen; the Cart Unit shows only what matters in store.

5. v3: multi-user and live, going physical (Weeks 10–11)

v3 is an **integration prototype** (Houde & Hill, 1997): it is real where the social experience depends on it. James built a live-sync backend and an Expo Cart Unit with **barcode scanning**; v0.3 added the household layer: requests, "Jay is shopping now", Ask home, two-phone Split Shop and a cost split. Prices are entered by hand, and compare orders products by **the shopper's** priority, never "healthier".



v3 Home Dock during a live shop, with a question from the store waiting.

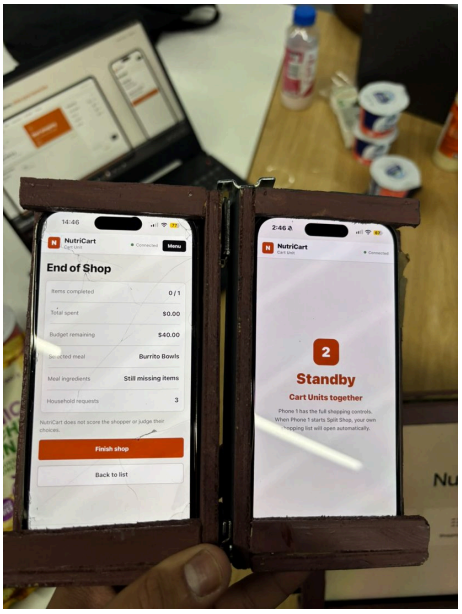


v3 Cart Units: Split Shop ("My half"), Ask home, neutral compare and the return summary.

James built physical housings for the Dock and the Cart Unit. Kenan's **app-v2** then redesigned the Hub and the Expo Cart Unit to match our combined concept; this is the version we tested in section 6 and showed at the tradeshow.



The tradeshow set-up: the Home Dock housing with two hinged Cart Unit mounts.



Split Shop on two Cart Units; the summary says NutriCart "does not score the shopper or judge their choices".

6. Evaluating with groups, in context (Week 12)

We ran two group sessions with four students who live together, one in their own kitchen (Evaluation Plan). All four valued the budget, Compare or deals most, and all struggled with first-time navigation. The sharpest finding: **the group chat is our real competitor**. G-01 called requests "another group chat" and felt the design was "promoting the opposite of being social", while G-02 valued seeing what was already at home. The social value sat in the live, in-store moment. So we will lead with same-time coordination (who's shopping, Ask home, live updates), work alongside the chat, simplify first use, and swap the LEDs for a chime.

7. Tradeshow (Week 12)



Teaching staff and visitors trying the Dock at our stall (faces blurred).

Over two days, visitors used the physical Dock and two Cart Units in pairs, and five teams left peer-feedback cards on Miro (summary). They confirmed the two-place design ("it splits the experience across two distinct physical environments", Team 18) and the poster's research-to-design strip. Three problems repeated:

- **First use is still hard.** Three teams asked for a short guide or simpler navigation, which confirms F2.
- **Unlabelled numbers.** Values in the meal editor had no units, and the deals tab and barcode compare were not fully working.
- **The group chat again.** "Instead of replacing group chats, integrate with it" (Team 18), which confirms F3.

Peers also asked whether the Hub makes housemates talk *less*. **Next iteration:** a first-use walkthrough, labels and units everywhere, a working compare, sharing to the group chat, and the v0.3 cost split with a private option (Ethics E1).

Limitations

Our samples were small, qualitative and mostly people we could reach, and we did not consult a domain expert such as a dietitian or student-housing staff; studio critique stood in. Early questions were sometimes framed as "healthier vs cheaper". Final testing was two 10–15 minute group sessions with four participants plus tradeshow visitors (peers, not real share houses), so it shows first use, not how a household behaves over weeks (Grudin, 1994); one session was screens only. v2 was Wizard of Oz, and v3 supports one household per server.

Who did what

Member	Main contributions
Aman Kushwaha (Project Management Lead)	Plan, charter, task board and stand-up records; synthesis; repo de-identification; evaluation plan and session G-02; pitch and tradeshow pack
James Olbes (Design & Prototyping Lead)	Shadowing session; physical prototypes (v2 cardboard, v3 housings); backend and Expo Cart Unit (v0.1–v0.2)

Member	Main contributions
Anujitha Fernando (Technical & Dev Lead)	GitHub repo set-up; integration and multi-device testing of v3; poster layout and printing
Kenan Chu (Documentation & Theory Lead)	Physical prototypes; interview protocol synthesis; personas and scenarios; research archive and README; app-v2 redesign (Hub + Expo Cart Unit); session G-01; frontend for Home hub and cart unit; WIKI and final PDF

We shared 23 research and testing sessions; the evidence register shows who ran each.

Use of AI: every member used Claude and ChatGPT; see the README for how, and how we checked the output.

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Ethical Considerations

We assessed NutriCart Hub with the **Ethical Disclaimer** from the Ethics for Designers toolkit (Gispen, 2017), because it asks designers to state who is affected, what could go wrong and what they take responsibility for, which suits a tool that sits inside a shared home. The disclaimer asks a team to describe the situation, list everyone affected, state its intentions, imagine how the design could cause unethical outcomes for each stakeholder, and decide openly which of those outcomes the team takes responsibility for. Following the course's emphasis on ethical design thinking throughout a project (Loke & Matthews, 2020), this is a **living page**: we revisit it at each milestone (section 6).

1. The current situation

Students in share houses near UQ St Lucia coordinate groceries informally: group chats, remembered requests, a list on someone's phone. Our research found price dominates their choices, "healthy" means different things to different people, and most people shop alone even when they live with others. Coordination problems are real (doubles, forgotten requests, awkward cost sharing), but so is sensitivity around money, food and bodies.

2. Who has a stake

Direct	Indirect
The housemate who shops (uses the Cart Unit)	Partners, guests and visitors who see the Dock in the kitchen
Housemates at home (use the Dock)	Supermarket staff and other shoppers (in-store use and testing)
A housemate on a tighter budget	Retailers whose prices appear in the product library
A housemate who doesn't share food or doesn't want to take part	Open Food Facts (receives barcode lookups)
Housemates with dietary, cultural, religious or allergy needs	Expo (the Cart Unit runs in the Expo Go app)
Someone with a difficult relationship with food or their body	Research participants and the team

3. Our intentions

NutriCart Hub is for **student share houses**. It is used **at the kitchen bench and in the supermarket aisle**. Its purpose is to **make the coordination housemates already do easier and fairer**: who needs what, who is shopping, what's already at home, and how to split the cost. It is not meant to change what anyone eats or to rate their choices.

The risks below are written for the full design (v0.3 in the repo). app-v2, the version we tested in Week 12, has a subset of its features: it has no cost split, so E1 and E5 apply when the split returns.

4. Imagining how it could go wrong

#	Stakeholder	How the design could harm them	What we changed or decided	Status
E1	Housemate on a tighter budget	The **cost split** and running total make income differences visible and turn shared shopping into a tally of debts. One participant said shared goals could be "stressful for one person because if he's getting less pay."	Shared staples are split evenly by default; personal requests are charged to whoever asked. Prices are optional, and unpriced items are left "to settle by hand". The summary can be dismissed without settling. *Next:* soften "owes" wording; let a household choose "don't split". **Tradeshow:** a peer worried that showing the budget and who owes what "might be a bit embarrassing for people who are tighter on money, and guests could see it too" (Team 18); another warned that seeing who spends what "could cause arguments". *Next:* a private settle-up shown only to the people involved, and hide totals when the Dock is idle.	Partly addressed; concern confirmed by peers
E2	Someone with a difficult relationship with food or their body	Group health goals and comparison can feel like judgement. A participant reacted to a dinner goal with: "It's essentially telling me that I'm fat... if anything, it could cause body dysmorphia." (Others found goals motivating, so reactions were mixed.)	v3 has **no group health goals, leaderboards, calories or health scores**. Compare is optional, shows facts only, and is ordered by the shopper's chosen priority. It never says "healthier".	Addressed by design
E3	Every housemate	**Awareness becomes monitoring** (Schmidt). Showing who added what, who is shopping and what they bought could feel like surveillance inside the home.	The Dock shows that someone is shopping and the store they chose. There is no live location tracking. Aliases are fine. **Honest limit:** past shops, their cost splits and barcode scans (with the shopper's alias) stay on the server until the next reset. *Next:* auto-delete past shops and scans after settling. **Tradeshow:** peers named spending and dietary preferences as what housemates may not want shared, and asked for control over what is shown; one team noted we already show "shopping activity without tracking users' locations" (Team 20). *Next:* per-item "just for me" privacy.	Partly addressed
E4	The housemate who ends up maintaining the Dock	**Unequal work**: one person does the adding and organising while everyone benefits (Grudin's work vs benefit disparity), and this invisible labour often falls on the same person.	Quick-add chips, a meal → "Add what's missing" button, requests that take one tap. Testing confirmed the upkeep feels "super tedious" (G01-P2); the finished shop now updates what's at home.	Confirmed in testing; partly addressed

#	Stakeholder	How the design could harm them	What we changed or decided	Status
E5	Housemates who don't share food, or don't want to take part	The design's **script** (Akrich) assumes a cooperative household; non-participants may feel pressured to join or be left out. **In v3, items marked "for the household" are split across every token, so a housemate who never uses NutriCart still appears as owing a share.**	Everything works for one person, and personal requests are only charged to whoever asked. *Next:* let a housemate opt out of shared costs, and remove unused tokens.	Partly addressed
E6	Housemates with dietary, cultural, religious or allergy needs	A product library and meal board built by our team reflect **our** food habits.	Products are editable in the Product library; compare shows the shopper's priority, not ours. Meals are currently fixed demo meals. *Next:* add and edit meals. *Next:* allergy and dietary notes per housemate visible on requests.	Open
E7	Housemates with low vision or colour-vision differences	Colour-only tokens exclude some users.	Tokens carry a **number and a name** as well as a colour; large touch targets. *Next:* contrast check.	Partly addressed
E8	Retailers, the household	**Commercial capture**: a compare screen is an obvious place to sell rankings to brands. Scraping prices could breach retailer terms.	Ordering is only by the shopper's chosen priority; no sponsored ordering, ever. Prices are entered by hand from observed shelf prices; **nothing is scraped**.	Decided
E9	Everyone using it	**Data**: shopping lists reveal diet, budget and routines. Barcode lookups go to Open Food Facts. We decided not to host the demo, so the database stays on our own laptop.	No accounts, no analytics, aliases only. The server stores list items, past shops with their cost splits, questions and barcode scans until the next reset. The demo database is reset between sessions. Database files are git-ignored.	Addressed for the prototype
E10	Other shoppers, store staff, research participants	**Research harms**: shadowing felt intrusive; filming in store captures bystanders.	Moved from shadowing to lower-burden sessions; consent script; manager permission in store, or a simulated shop; no faces or bystanders; recordings and transcripts kept off public GitHub, deleted at semester end.	Addressed
E11	The shopper	**Dependence**: if the server or connection fails mid-shop, the list disappears.	Automatic fallback from live connection to polling; hotspot setup. **Tradeshow:** a peer described poor reception in the supermarket leading to an item bought twice and a wrong budget (Team 16). *Next:* keep a cached copy on the Cart Unit and show when it is out of date.	Partly addressed

5. What we take responsibility for, and what we don't

We take responsibility for: E1–E11 as far as the design can reach. These are consequences of choices we made (what's shown, to whom, how costs are split, what's stored, how we ran research). Winner (1980) reminds us that a detail like "split shared items evenly" quietly settles a social question for the

household. So the split rule is visible and explained, not hidden, and making it optional is our next step (E1).

Outside our control, but named: existing conflict between housemates (the Dock can surface it but not resolve it); retailer pricing accuracy; a household choosing to use the cost split to pressure someone. We can't prevent these, so we don't design features that amplify them, such as rankings, reminders to pay, or debt history.

6. Revision log

Date	Milestone	What changed on this page
Wk 5	Proposal Q&A and peer comments	Q&A raised: what is "healthy" for everyone; dietary, religious and cultural needs; eating disorders. Peer comments raised privacy and user control.
Wk 7–8	Prototype interviews (v1, v2)	Evidence for E1 (pay) and E2 (body image) → group goals made optional; later left out of v3
Wk 8	Shadowing attempt	E10: method changed
13 Oct	v3 build	E3, E8, E9, E11 added; first full disclaimer
14 Oct	Research archive	Transcripts removed from the repo and its history; all sessions re-published as de-identified records; raw files kept privately on OneDrive (E10). Lesson: de-identify before anything leaves the private folder.
19 Oct	Evaluation round 1 (G-01, G-02)	**E4 confirmed:** keeping Already at home updated "seems super tedious" (G01-P2), so the finished shop updates it automatically (v0.3) and one-line quick-add is next. **E5 strengthened:** G-01 would rather use the group chat, so NutriCart must work alongside the chat instead of pulling the whole house onto it. **E1 and E3 not yet tested** (app-v2 has no cost split; the privacy question wasn't recorded), so we ask tradeshow visitors.
20–22 Oct	Tradeshow	Peer feedback from five teams (summary). **E1 confirmed** (visible budgets and debts could embarrass a housemate on less money, or be seen by guests). **E3 refined** (spending and dietary preferences are the sensitive data; location is not shown). **E5 echoed** (a housemate who won't use the Hub; one person always paying). **E11 confirmed** (poor reception in store). Next steps added to each row.

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